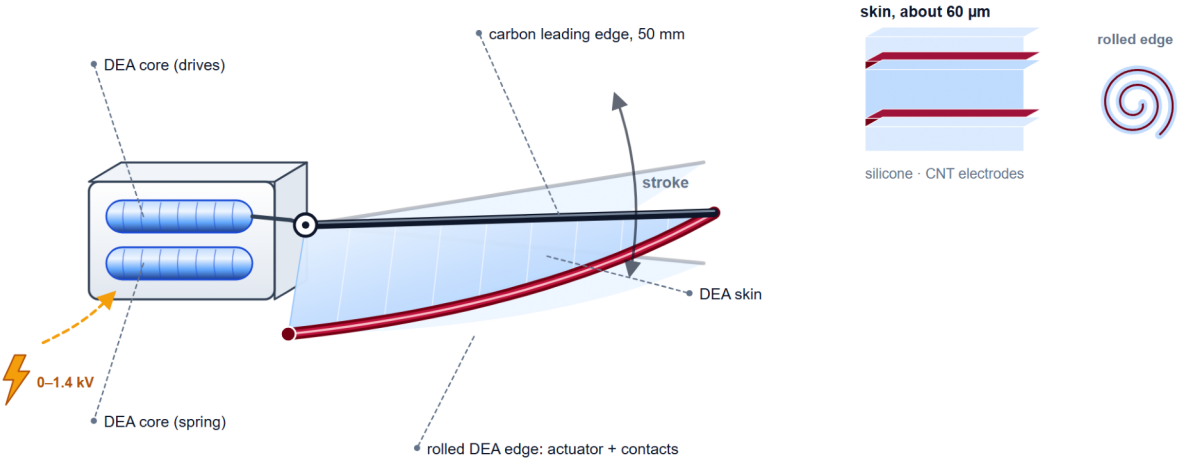
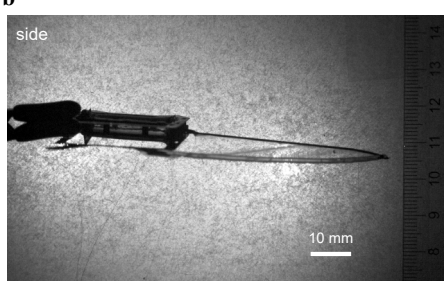
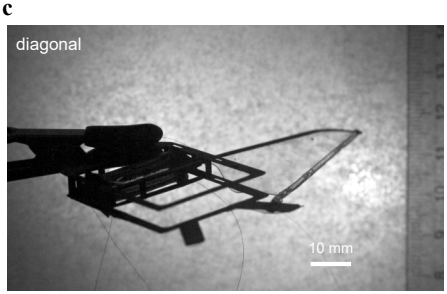
a


DEA core (drives)
carbon leading edge, 50 mm
stroke
DEA skin
0–1.4 kV
DEA core (spring)
rolled DEA edge: actuator + contacts
skin, about 60 µm
rolled edge
silicone · CNT electrodes

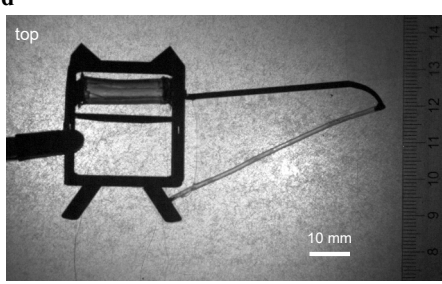
b


side
10 mm

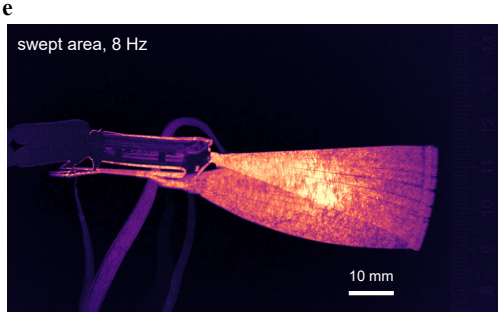
c


diagonal
10 mm

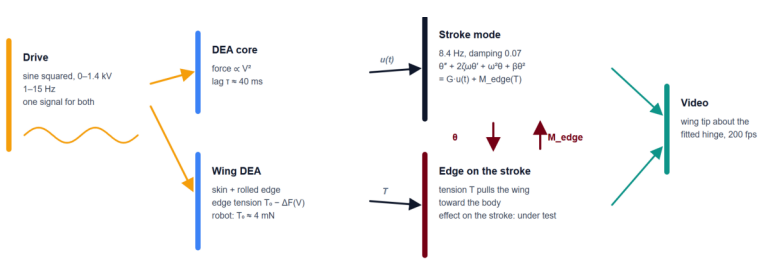
d


top
10 mm

e


swept area, 8 Hz
10 mm

f


Drive
sine squared, 0–1.4 kV
1–15 Hz
one signal for both
DEA core
force ∝ V²
lag τ ≈ 40 ms
Wing DEA
skin + rolled edge
edge tension T₀ − ΔF(V)
robot: T₀ ≈ 4 mN
u(t)
Stroke mode
8.4 Hz, damping 0.07
θ'' + 2ζω₀θ' + ω₀²θ + βθ³
= G·u(t) + M_edge(T)
θ
M_edge
T
Edge on the stroke
tension T pulls the wing
toward the body
effect on the stroke: under test
Video
wing tip about the
fitted hinge, 200 fps
one drive reaches the stroke through the core and, possibly, through the wing's tension